# MATHEMATICAL FOUNDATIONS FOR CLUSTERING, SIMILARITY, AND NETWORK ANALYSIS

CONTENTS

Two kinds of mathematical frameworks are discussed in this chapter: clustering algorithms, and metrics for evaluating similarity and clustering performance.

## 1 LINEAR ALGEBRA DISTANCE CALCULATIONS

## 2 SIMILARITY METRICS

### 2.1 *Tanimoto Coefficient (also known as the Jaccard Index)*

**Range:** 0-1 **Higher is More Similar:** True

We define the Tanimoto Coefficient (TC) between two sets (A, B) as such:

$$TC(A, B) = TC(B, A) = \frac{|A \cap B|}{|A \cup B|} = \frac{|A \cap B|}{|A| + |B| - |A \cap B|}$$

The related Jaccard Distance is simply $1 - TC$.

This is a favorite for comparing bitmaps (i.e. chemical fingerprints), in part because calculating the intersection and union of two bitvectors is very efficient.

As an example, given ligand A and ligand B with fingerprints:

$$A = |0|1|0|1|0|$$

$$B = |0|1|1|0|0|$$

$$TC(A, B) = \frac{|0|1|0|0|0|}{|0|1|1|1|0|} = \frac{1}{3} = 0.33$$

### 2.2 *Dice-Sørensen Coefficient*

**Range:** 0-1 **Higher is More Similar:** True

Following the conventions above, the Dice-Sørensen Coefficient can be calculated by:

$$DC(A, B) = \frac{2|A \cap B|}{|A| + |B|}$$

Although similar to the Tanimoto Coefficient, this function does not satisfy the triangle inequality and is much less frequently used.

### 2.3 *Tversky Index*

**Range:** 0-1 **Higher is More Similar:** True

The Tversky Index is a generalization of the Tanimoto Coefficient and Dice-Sørensen Coefficient, and is defined as:

$$TI(A, B) = \frac{|A \cap B|}{\alpha|A \backslash B| + \beta|B \backslash A| + |A \cap B|}$$

Note that the Tanimoto Coefficient is a special case of the Tversky Index where $\alpha = \beta = 1$, and the Dice-Sørensen Coefficient is a special case where $\alpha = \beta = 0.5$. Although $\alpha$ and $\beta$ need not sum to 1, they are of special interest. If $\alpha \neq \beta$, the Tversky Index is asymmetric, and the value of $\alpha$ and $\beta$ can be tuned to emphasize the importance of the first or second set.

### 2.4 *Cosine Similarity*

## 3 CLUSTERING METRICS

#### 3.0.1 Silhouette Coefficient

The silhouette coefficient is a measure of how similar an object is to its own cluster compared to other clusters. First we calculate the mean intra-cluster distance $a_i$ by taking the average of each distance $d(i, j)$ for each object $i$ in cluster $C_i$:

$$a_i = \frac{1}{|C_I| - 1} \sum_{j \in C_I, i \neq j} d(i, j)$$

Next we calculate the mean nearest-cluster distance $b_i$ for each object $i$ in cluster $C_i$:

$$b_i = \min_{J \neq I} \frac{1}{|C_J|} \sum_{j \in C_J} d(i, j)$$

Then we can define the silhouette coefficient is defined for each object $i$ as:

$$SC_i = \frac{b_i - a_i}{\max(a_i, b_i)}$$

Finally, by taking the mean silhouette coefficient over all objects, the final value will be between -1 and 1, where a high value indicates a good clustering.